Waste collection routing problem with stochastic time-dependent demand increment

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1 Introduction and Motivation

Disposal activities are fundamental for the environment and the citizens. In particular, the EU rules related to waste management and the emissions generated by collecting and processing huge amounts of end-of-life products call attention to local recovery networks. In many EU countries, the collection of household waste and material is carried out via a combination of different systems (door-to-door modality, pick-up from common containers/areas, etc.). Generally, the daily collection refers to specific recyclable waste, e.g. glass or paper, and a separate collection problem is solved for each waste flow. Since the service can be carried out at the collection points (CPs) along the whole day, the collected amount strongly depends on the part of the day on which each CP is visited. For example, a restaurant does not produce waste in the morning, since it is closed, while the largest amount is produced after the lunch services. On the contrary, a school produces waste only during the morning since it is closed in the afternoon, and so on. Hence, if a CP is visited in the first part of the day, residual waste will not be collected until the next day. Although information about expected waste in different parts of the day can be derived from historical data, current practice usually ignores it. The goal of this work is to study how information and demand forecast could be effectively exploited to reduce costs for the companies and to offer a better service for the community by avoiding letting cumulus of uncollected waste for a long time.

More specifically, in our problem a set of CPs must be visited by a set of identical capacitated vehicles located at a common depot. Each CP has a known initial quantity of recyclable waste which may increase along the day. We divide the day in not-overlapping time-slots. For each CP and each time-slot, the uncertain demand increment is represented as a random variable, taking values

within a finite interval. Since the increment depends on the time-slot, time-dependent demands arise and our problem is referred to as Waste Collection Routing Problem with Stochastic Time-Dependent Demand Increment (WCRP-STDDI). We assume that the vehicle loads all the waste if its capacity is sufficient; otherwise, it partially collects the waste (up to its capacity) or totally skips the CP. The goal is to minimize the expected total cost given by net routing costs and penalties associated with uncollected waste at the end of the day.

Although waste collection routing problems with stochastic demand have been extensively studied in the literature, most works consider the demand unknown but stationary for the whole planning horizon. Instead Markov et al. [1] deal with a demand varying on time. However, their problem strongly differs from ours since they consider a multi-period setting in which demands may increase from one period to another, but it is constant over the same period. This means that, since the collection routes span on a single period, the exact quantity to collect is known prior to make the daily plan. Conversely, we study a single day setting, where the time-horizon is split in time-slots and routes can span over multiple time-slots. Therefore, since the demand may very across time-slots, we do not have a deterministic information about the quantity to collect, when the route are constructed and start. Hence, the collection plan may turn out to be infeasible and a recourse action, consisting in partially emptying a CP or in skipping one or more CPs, may be needed.

2 Solution Approach and Preliminary Results

The WCRP-STDDI is modeled as a two-stage stochastic programming (SP) model in which the first-stage decisions refer to the collection plan, whereas recourse actions are applied at the second level when the actual amount of waste at each CP is revealed. We provide a mathematical formulation for the problem and solve it via the scenario sampling analysis. We compare the results of the SP model with those achievable with different deterministic policies. Furthermore, we derive interesting managerial insights on how to derive effective policies to construct the collection plan. Preliminary results show that neglecting uncertainty yields to poor results.

References

[1] I. Markov, M. Bierlaire, J.-F. Cordeau, Y. Maknoon, S. Varone, "Waste collection inventory routing with non-stationary stochastic demands", *Computers & Operations Research* 113, 104798 (2020).